



If I Fits,
I Harness Sits

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CCat Leash Harness

Project PDF

A production-ready CCAT practice platform with a full Next.js/Postgres web application, timed quiz workflow, analytics, and a governed AI question-generation lab.

LIVE WEB APP	https://quizcat-web-production.up.railway.app
REPOSITORY	https://github.com/micronox/FiredFest_Submission
BUILDER	Larry Vallely / Blace Houle
SUBMISSION	Fired Festival

If I Fits, I Harness Sits

A beautiful little engine for disciplined practice, measurable progress, and safe question generation.

Project Snapshot

CCat Leash Harness converts a terminal-first CCAT trainer into a deployed browser application. It preserves a carefully structured 400-question seed bank, adds persistent attempts and statistics, and introduces Question Lab: an AI generation workflow that never writes directly to production data without deterministic review.

400**Seed questions**

Imported from eight CCAT-style sample exams.

1,952**Choices**

Preserved labels, order, answer key, and explanations.

8**Base tests**

Fifty-question exam forms with 15-minute timers.

6**Core tables**

Questions, choices, tests, mappings, attempts, answers.

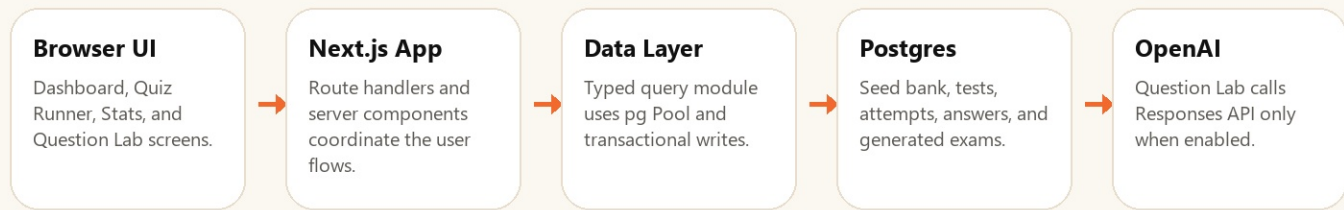
What the product does

- Lets users launch full source exams or create generated practice tests from the validated bank.
- Runs timed quizzes with answer persistence, immediate correctness checks, explanations, and finish states.
- Tracks completed attempts for a stats screen that makes progress visible after every run.
- Provides an LLM-powered Question Lab gated by schema validation, similarity checks, safety checks, and deterministic arithmetic verification.

Design thesis

The application is built around a simple promise: practice should feel focused, measurable, and trustworthy. The interface is intentionally direct, while the back end keeps the source dataset, quiz sessions, scoring, and AI-generated candidates auditable.

System Architecture



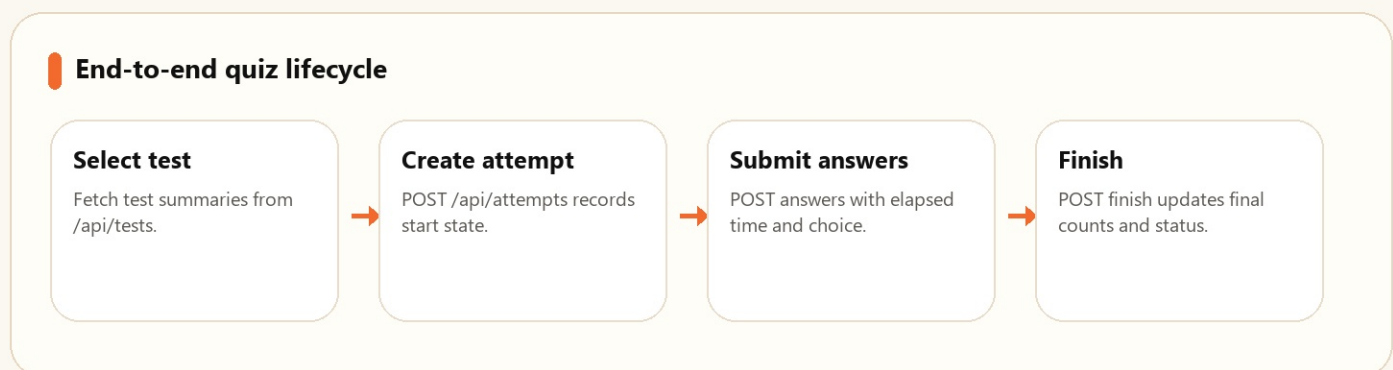
Runtime components

- Next.js 16 / React 19 application under web/.
- Postgres schema in web/db/schema.sql.
- CSV migration and seed scripts preserve the original question bank.
- Static image stimuli are served from web/public/images/.
- Railway deploys web/ as the service root with npm run start.

Operational boundaries

- No auth in the practice app; intended for public demo and personal use.
- Answer keys stay server-side until a question is answered.
- Question Lab is disabled unless OPENAI_API_KEY and QUESTION_LAB_ENABLED are set.
- The access token creates an HttpOnly session rather than exposing secrets client-side.
- Generated candidates are review-only and never auto-inserted into the bank.

End-to-end quiz lifecycle



Complete Database Schema

The schema separates reusable question content from test assembly and user performance. This keeps the imported CCAT bank immutable in spirit while allowing randomized generated exams and persistent attempt history.

Table	Column	Type	Purpose
questions	id	SERIAL PK	Primary question identifier.
questions	external_id	TEXT UNIQUE	Stable imported source id.
questions	origin	TEXT NOT NULL	Import origin / source family.
questions	source_exam	TEXT	Exam grouping from seed data.
questions	source_file	TEXT	Original source file reference.
questions	source_category	TEXT	Raw imported category.
questions	source_question_number	TEXT	Question number in source.
questions	category	TEXT NOT NULL	Normalized broad category.
questions	question_type	TEXT NOT NULL	Normalized skill type.
questions	prompt	TEXT NOT NULL DEFAULT "	Reusable instruction text.
questions	stimulus	TEXT NOT NULL	Question-specific text, table, or image filename.
questions	stimulus_type	TEXT CHECK	One of text, text_table, image.
questions	correct_choice_label	TEXT NOT NULL	Correct answer label.
questions	correct_choice_text	TEXT NOT NULL	Correct answer text.
questions	explanation	TEXT NOT NULL DEFAULT "	Plain-text rationale.

Table	Column	Type	Purpose
choices	id	SERIAL PK	Choice identifier.
choices	question_id	INTEGER FK -> questions.id ON DELETE CASCADE	Parent question.
choices	label	TEXT NOT NULL	A-D label preserved from source.
choices	position	INTEGER NOT NULL	Rendered ordering.
choices	text	TEXT NOT NULL	Choice body.
choices	UNIQUE	(question_id, label), (question_id, position)	Prevents duplicate labels or order slots.
tests	id	SERIAL PK	Test identifier.
tests	title	TEXT NOT NULL	Exam or generated-test title.
tests	kind	TEXT NOT NULL	Seed/source exam or generated practice type.
tests	source_exam	TEXT UNIQUE	Original exam id for seeded tests.
tests	question_count	INTEGER NOT NULL	Number of assigned questions.
tests	time_limit_seconds	INTEGER DEFAULT 900	Timer limit; generated tests use one minute per question.
test_questions	test_id	INTEGER FK -> tests.id ON DELETE CASCADE	Parent test.
test_questions	question_id	INTEGER FK -> questions.id ON DELETE CASCADE	Assigned question.
CCat Leash Harness test_questions	position	INTEGER NOT NULL	Position inside the test.

Complete Database Schema

Table	Column	Type	Purpose
attempts	id	SERIAL PK	Attempt identifier.
attempts	test_id	INTEGER FK -> tests.id	Attempted test.
attempts	status	TEXT CHECK	in_progress, completed, timed_out, or aborted.
attempts	started_at	TIMESTAMPTZ NOT NULL	Server start timestamp.
attempts	finished_at	TIMESTAMPTZ	Server finish timestamp.
attempts	elapsed_seconds	DOUBLE PRECISION DEFAULT 0	Current or final elapsed time.
attempts	answered_count	INTEGER DEFAULT 0	Stored denormalized answered count.
attempts	correct_count	INTEGER DEFAULT 0	Stored denormalized correct count.
attempts	total_questions	INTEGER NOT NULL	Total expected questions.
attempt_answers	id	SERIAL PK	Answer identifier.
attempt_answers	attempt_id	INTEGER FK -> attempts.id ON DELETE CASCADE	Parent attempt.
attempt_answers	question_id	INTEGER FK -> questions.id	Answered question.
attempt_answers	question_position	INTEGER NOT NULL	Position in test at answer time.
attempt_answers	selected_choice_label	TEXT NOT NULL	Selected label.
attempt_answers	selected_choice_text	TEXT NOT NULL	Selected answer body.
attempt_answers	is_correct	BOOLEAN NOT NULL	Server-scored correctness.
attempt_answers	elapsed_seconds	DOUBLE PRECISION NOT NULL	Elapsed time when answer was submitted.
attempt_answers	UNIQUE	(attempt_id, question_id), (attempt_id, question_position)	One answer per question and position.

Indexes

```

idx_choices_question_position      ON choices(question_id, position)

idx_test_questions_position        ON test_questions(test_id, position)

idx_attempt_answers_attempt        ON attempt_answers(attempt_id, question_position)

```

Design note: relational constraints preserve answer ordering and scoring integrity across retries.

API Contract

Method	Route	Responsibility
GET	/api/tests	List seed and generated tests.
POST	/api/tests	Create a generated 8-10 question practice test.
GET	/api/tests/[id]	Fetch a test definition with client-safe questions and choices.
POST	/api/attempts	Create an in-progress attempt for a selected test.
GET	/api/attempts/[id]	Read current attempt state.
POST	/api/attempts/[id]/answers	Record or update an answer; return answer key, explanation, and correctness.
POST	/api/attempts/[id]/finish	Finish, time out, or abort an attempt and calculate result summary.
POST	/api/question-lab	Generate a governed candidate question when the lab is enabled and authorized.

TypeScript domain schema

- ClientQuestion: id, externalId, category, questionType, prompt, stimulus, stimulusType, choices.
- Question: server-side extension with origin, source metadata, correct choice, and explanation.
- TestSummary/TestDefinition: title, kind, source exam, question count, time limit, and questions.
- QuizAttempt: status, timestamps, elapsed seconds, answered count, correct count, total.
- AttemptAnswer/AnswerResult: selected choice, correctness, elapsed time, answer key, explanation.
- QuizResult/AttemptSummary: final scoring and history rows for the stats screen.

Privacy and scoring posture

The client receives no answer key in the test payload. Scoring happens server-side when an answer is submitted, then the API returns correctness, explanation, and the correct answer for immediate learning feedback.

Question Lab Governance

Question Lab is a controlled generation loop. It asks a model for one original CCAT-style candidate, validates it against deterministic checkpoints, and gives the model one revision pass with structured feedback. Accepted candidates remain review artifacts, not automatic production inserts.

Schema object	Fields
CandidateQuestion	category, questionType, prompt, stimulus, choices, correctChoiceLabel, explanation, verificationExpression
CandidateChoice	label, text
HarnessCheckpoint	name, passed, detail
HarnessAttempt	revision, candidate, checkpoints
HarnessResult	accepted, model, attempts, candidate, checkpoints

Checkpoint battery

Schema

Required fields plus exactly four choices.

Similarity

Highest seed-question token similarity must stay under 0.72.

Single answer

Unique A-D labels and one declared correct choice.

Math

Arithmetic types must pass a safe deterministic calculator.

Explanation

Substantive rationale that names the correct choice text.

Content safety

Basic study-tool screen for unsafe content.

Category

Question type must exactly match the requested seed type.

Export readiness

Candidate must serialize cleanly for review/export.

Model boundary

The model is configured through OPENAI_MODEL, defaults to gpt-4o-mini, has a 20-second timeout, zero SDK retries, a 1,200-token cap, and a strict Zod response schema. Public generation is rate-limited by session and request budget.

Experience + Operations

User-facing screens

Dashboard

Choose source exams, create generated practice, launch Question Lab.



Quiz

Timed practice, persistent answers, explanations, and result state.



Stats

Completed attempts sorted by finish time with scoring summary.



Lab

Generate governed candidates and inspect checkpoint feedback.

Deployment

- Railway service: quizcat-web.
- Start command: npm run start.
- Service root: web/ with path-as-root deployment.
- Required database variable: POSTGRES_URL.
- Production seed state: 400 questions, 1,952 choices, 8 tests.

Environment switches

POSTGRES_URL

OPENAI_API_KEY

OPENAI_MODEL

QUESTION_LAB_ENABLED

QUESTION_LAB_ACCESS_TOKEN

Submission links

PRODUCTION	https://quizcat-web-production.up.railway.app
REPOSITORY	https://github.com/micronox/FiredFest_Submission
TEXTUAL BROWSER APP	https://quizcat-textual-production.up.railway.app



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Built to make practice measurable.

CCat Leash Harness is more than a quiz UI: it is a schema-respecting practice system, a deployed web product, and a controlled AI workflow for expanding the bank without losing trust in the answers.

Core stack

Next.js, React, TypeScript, Postgres, Railway

Data integrity

Relational constraints, ordered choices, server-side answer keys

AI safety

Typed schema, deterministic checkpoints, similarity limit, calculator verification

Reader promise

Fast practice, clear feedback, visible progress

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